

Package ‘PathoFMPred’

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Type Package

Title Cancer-Specific Molecular Prediction from Pathology Foundation Models

Version 0.3.0

Description Builds and applies portable cancer-specific partial least-squares regression and PLS-LDA model collections from pathology foundation-model representations. Feature and outcome tables are linked by explicit patient identifiers, repeated slide rows are pooled at patient level, and model development uses nested cross-validation before final fitting. The public package includes only a minimal Giga-SSL example; optional full Giga-SSL and Prov-GigaPath collections are obtained by an explicit user download. TITAN-derived fitted parameters are not distributed publicly. Outputs are research estimates without external-validation or clinical-use claims.

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Encoding UTF-8

Depends R (>= 4.2.0)

Imports fastPLS,
ggplot2,
rmarkdown,
stats,
utils

Suggests knitr,
testthat (>= 3.0.0)

VignetteBuilder knitr

Config/testthat/edition 3

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available_foundation_models
<i>List foundation-model representations bundled with the package</i>

Description

List foundation-model representations bundled with the package

Usage

available_foundation_models()

Value

A character vector. Availability means that at least one fitted research model is bundled; it does not imply external validation.

available_pathofm_models
<i>List available cancer-specific models</i>

Description

List available cancer-specific models

Usage

```
available_pathofm_models(  
  foundation_model = NULL,  
  cancer = NULL,  
  outcome_type = NULL,  
  family = NULL,  
  include_limited_evidence = FALSE  
)  
  
available_titan_models(  
  cancer = NULL,  
  outcome_type = NULL,  
  family = NULL,  
  include_limited_evidence = FALSE  
)
```

Arguments

foundation_model	One or more of ‘TITAN’, ‘GigaSSL’, or ‘ProvGigaPath’.
cancer	Optional TCGA cancer code or vector of codes.
outcome_type	Optional ‘continuous’ or ‘binary’ filter.
family	Optional endpoint-family filter.
include_limited_evidence	Include limited-sample models only when they crossed the representation-specific matched effect threshold. Models tested below the threshold never enter prediction or reporting. The default is ‘FALSE’.

Value

A filtered model-registry data frame.

compare_pathofm_models

Compare representation-specific models for one cancer-endpoint task

Description

Lists controlled reference objects for the same task while distinguishing object availability from representation-specific matched-atlas effect-threshold crossing. The inventory is a partial operationalization of the multi-representation atlas. The function does not automatically select the largest internal estimate or combine representation-specific predictions.

Usage

```
compare_pathofm_models(cancer, endpoint, outcome_type = NULL, family = NULL,
  include_limited_evidence = FALSE)
```

Arguments

cancer	TCGA cancer code.
endpoint	Exact endpoint label.
outcome_type	Optional continuous or binary filter.
family	Optional endpoint-family filter.
include_limited_evidence	Include limited-sample models only when they crossed the representation-specific matched effect threshold.

Value

A registry data frame with object-availability, matched-threshold, resource-scope and selection-guidance fields.

```
create_pathofmpred_object
```

Build a portable PathoFMPred model collection

Description

Feature and outcome tables are linked only through 'id_column'. Outcome IDs must be unique. Feature IDs may repeat, for example when a patient has more than one slide, and are pooled before model development.

Usage

```
create_pathofmpred_object(
  feature_table,
  outcome_table,
  id_column,
  aggregation = c("mean", "median"),
  foundation_model = "custom",
  feature_columns = NULL,
  outcome_columns = NULL,
  outcome_types = NULL,
  control = pathofmpred_control(),
  output_file = NULL
)
```

Arguments

feature_table	Data frame, matrix, or CSV path containing identifiers and numeric foundation-model features.
outcome_table	Data frame or CSV path containing one unique row per identifier and one or more outcomes.
id_column	Name of the identifier column present in both tables.
aggregation	How repeated feature rows are pooled: 'mean' or 'median'.
foundation_model	Label stored in the resulting object.
feature_columns	Optional numeric feature columns. By default all columns except 'id_column' are used.
outcome_columns	Optional outcomes to fit. By default all columns except 'id_column' are considered.
outcome_types	Optional named character vector containing 'continuous' or 'binary'. Otherwise numeric two-level outcomes are binary and other numeric outcomes are continuous.
control	Settings created by [pathofmpred_control()].
output_file	Optional '.rds' destination.

Value

A ‘PathoFMPredObject’ containing fitted models, a registry, nested-CV summaries, and a linkage/missingness audit.

```
example_pathofm_features
```

Construct a synthetic feature vector for examples and smoke tests

Description

The vector is the across-model mean of stored training-feature means for the requested cancer and is not a patient record.

Usage

```
example_pathofm_features(
  cancer = "BRCA",
  foundation_model = "TITAN",
  include_limited_evidence = FALSE
)

example_titan_features(cancer = "BRCA", include_limited_evidence = FALSE)
```

Arguments

cancer	A cancer code with bundled models.
foundation_model	One of ‘TITAN’, ‘GigaSSL’, or ‘ProvGigaPath’.
include_limited_evidence	Use limited-evidence models when constructing the synthetic across-model feature mean. The default is ‘FALSE’.

Value

A one-row data frame containing the named features required by the selected representation.

```
fetch_pathofmpred_models
```

Download optional full PathoFMPred model collections

Description

Downloads occur only after an explicit user call. The public package does not download model files during installation, loading, checking, or vignette construction. TITAN-derived fitted parameters are deliberately unavailable through this function.

Usage

```

fetch_pathofmpred_models(
  foundation_model = c("GigaSSL", "ProvGigaPath"),
  destination = .pathofm_user_model_dir(),
  base_url = getOption("PathoFMPred.model_base_url",
    "https://github.com/tkcaccia/PathoFMPred/releases/download/models-v1"),
  overwrite = FALSE,
  quiet = FALSE
)

```

Arguments

foundation_model	One or both of ‘GigaSSL’ and ‘ProvGigaPath’.
destination	Directory in which model collections will be stored.
base_url	Release directory containing the model files. Advanced users may point this at an institutional mirror.
overwrite	Replace an existing file.
quiet	Passed to [utils::download.file()].

Value

Named paths to the downloaded model collections.

pathofmpred_control	<i>Define model-development settings</i>
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Description

Define model-development settings

Usage

```

pathofmpred_control(
  components = 1:20,
  outer_folds = 5L,
  inner_folds = 5L,
  repeats = 1L,
  scaling = "centering",
  method = "plssvd",
  backend = NULL,
  binary_selection = "AUROC",
  continuous_selection = "auto",
  seed = 1L,
  minimum_continuous = 50L,
  minimum_binary_class = 20L,
  rsvd_oversample = NULL,
  rsvd_power = NULL
)

```

Arguments

components	Candidate PLS component counts.
outer_folds, inner_folds	Numbers of nested cross-validation folds.
repeats	Number of independently seeded nested evaluations.
scaling, method, backend	Arguments passed to fastPLS.
binary_selection	Inner-CV objective for binary outcomes. AUROC is the default because the resulting score ranking is independent of prevalence- specific class priors.
continuous_selection	Inner-CV objective for continuous outcomes.
seed	Initial random seed.
minimum_continuous	Minimum labelled patients for a continuous model.
minimum_binary_class	Minimum patients in each binary class.
rsvd_oversample, rsvd_power	Optional overrides. 'NULL' preserves the installed fastPLS defaults.

Value

A settings list for [create_pathofmpred_object()].

pathofm_model_registry

Read the bundled pathology-foundation-model registry

Description

Read the bundled pathology-foundation-model registry

Usage

```
pathofm_model_registry()
```

```
titan_model_registry()
```

Value

A data frame with one row per fitted cancer-endpoint model.

pathofm_sample_report *Create a single-sample pathology-foundation-model research output*

Description

Create a single-sample pathology-foundation-model research output

Usage

```
pathofm_sample_report(
  cancer,
  features,
  foundation_model = c("TITAN", "GigaSSL", "ProvGigaPath"),
  patient_id = NULL,
  output_file = "pathofm_prediction_report",
  format = c("html", "pdf", "both"),
  clinical_context = NULL,
  include_limited_evidence = FALSE,
  quiet = TRUE
)

titan_sample_report(
  cancer,
  features,
  patient_id = NULL,
  output_file = "titan_prediction_report",
  format = c("html", "pdf", "both"),
  clinical_context = NULL,
  include_limited_evidence = FALSE,
  quiet = TRUE
)
```

Arguments

cancer	A single TCGA cancer code.
features	One or more slide-feature rows for one patient.
foundation_model	One of 'TITAN', 'GigaSSL', or 'ProvGigaPath'.
patient_id	Optional patient/sample identifier for all rows.
output_file	File stem or '.html'/'pdf' path.
format	One of 'html', 'pdf', or 'both'.
clinical_context	Optional named list with 'shared_features', 'pathology', and 'source' character fields. This information is displayed as supplied and is not used by any prediction model.
include_limited_evidence	Explicitly opt in to limited-sample models that still crossed the matched effect threshold for the selected cancer and representation. Tested-below-threshold objects never enter reports.
quiet	Passed to [rmarkdown::render()].

Value

Invisibly, the generated file path(s).

plot_pathofm_binary	<i>Plot PathoFMPred binary outputs</i>
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Description

Representation-neutral alias for [plot_titan_binary()]. Binary score ranks remain explicitly uncalibrated ranks, not probabilities.

Usage

```
plot_pathofm_binary(predictions)
```

Arguments

predictions Output from [predict_pathofm()] for one patient.

Value

A ggplot object.

plot_pathofm_radar	<i>Radar plot of PathoFMPred continuous outputs</i>
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Description

Representation-neutral alias for [plot_titan_radar()]. The selected foundation model is carried in the prediction rows returned by [predict_pathofm()]; no representation is selected inside the plotting function.

Usage

```
plot_pathofm_radar(predictions, max_endpoints = 12L)
```

Arguments

predictions Output from [predict_titan()] for one patient.
max_endpoints Maximum number of endpoints, prioritising extremes.

Value

A ggplot object.

plot_titan_binary	<i>Plot binary PLS-LDA outputs</i>
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Description

Plot binary PLS-LDA outputs

Usage

```
plot_titan_binary(predictions)
```

Arguments

predictions Output from [predict_titan()] for one patient.

Value

A ggplot object. Scores are displayed as TCGA out-of-fold score ranks explicitly labelled "not probability".

plot_titan_radar	<i>Radar plot of continuous prediction percentiles</i>
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Description

Radar plot of continuous prediction percentiles

Usage

```
plot_titan_radar(predictions, max_endpoints = 12L)
```

Arguments

predictions Output from [predict_titan()] for one patient.
max_endpoints Maximum number of endpoints, prioritising extremes.

Value

A ggplot object. Endpoint names and original model predictions are placed at the outer radar corners to avoid overlap with the profile.

predict_pathofm	<i>Apply supported models for one pathology foundation-model representation</i>
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Description

Multiple rows with the same ‘patient_id’ are interpreted as multiple slides and mean-pooled before prediction, matching model development.

Usage

```
predict_pathofm(
  cancer,
  features,
  foundation_model = c("TITAN", "GigaSSL", "ProvGigaPath"),
  patient_id = NULL,
  outcome_type = c("continuous", "binary"),
  family = NULL,
  warn_ood = TRUE,
  include_limited_evidence = FALSE
)
```

Arguments

cancer	One TCGA cancer code per row, or one code recycled to all rows.
features	A matrix or data frame containing the correctly named features for ‘foundation_model’.
foundation_model	One of ‘TITAN’, ‘GigaSSL’, or ‘ProvGigaPath’.
patient_id	Optional identifier per row. Repeated identifiers are pooled.
outcome_type	One or more of ‘continuous’ and ‘binary’.
family	Optional endpoint-family filter.
warn_ood	Warn when over 5 percent of dimensions fall outside training ranges.
include_limited_evidence	Explicitly opt in to limited-sample models that still crossed the representation-specific matched effect threshold. Models tested below the threshold never enter prediction or reporting.

Value

A long data frame with one row per patient and applicable model.

predict_titan	<i>Apply the legacy TITAN model interface</i>
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Description

Backward-compatible wrapper for [predict_pathofm()] with ‘foundation_model = "TITAN"‘.

Usage

```
predict_titan(
  cancer,
  features,
  patient_id = NULL,
  outcome_type = c("continuous", "binary"),
  family = NULL,
  warn_ood = TRUE,
  include_limited_evidence = FALSE
)
```

Arguments

cancer	One TCGA cancer code per row, or one code recycled to all rows.
features	A matrix or data frame containing the correctly named features for ‘foundation_model‘.
patient_id	Optional identifier per row. Repeated identifiers are pooled.
outcome_type	One or more of ‘continuous‘ and ‘binary‘.
family	Optional endpoint-family filter.
warn_ood	Warn when over 5 percent of dimensions fall outside training ranges.
include_limited_evidence	Explicitly opt in to limited-sample models that still crossed the representation-specific matched effect threshold. Models tested below the threshold never enter prediction or reporting.

validate_pathofmpred_object	<i>Validate a PathoFMPred model collection</i>
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Description

Validate a PathoFMPred model collection

Usage

```
validate_pathofmpred_object(object)
```

Arguments

object	An object created by [create_pathofmpred_object()] or a distributed PathoFMPred model collection.
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Value

‘TRUE’, invisibly, when the object is valid.

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